Implementation Budget / Cost Specification

The following budget covers one development laptop, one Android phone, one Galaxy Watch6, one month of internet and electricity, and a project food allowance. Amounts are expressed in Philippine pesos (PHP).

Item	Cost Basis	Amount
Laptop	One unit	₱30,000.00
Android phone	One unit	₱14,000.00
Samsung Galaxy Watch6	One unit	₱4,500.00
Hardware subtotal		**₱48,500.00**
Visual Studio Code and Android Studio	No-cost development tools	₱0.00
React Native, Expo and required SDKs	Prototype development	₱0.00
Supabase	Free-plan allowance	₱0.00
Firebase FCM and Expo Push Service	No-cost messaging services	₱0.00
Gemini API and Expo build services	Within available no-cost quotas	₱0.00
Software subtotal		**₱0.00**
Internet connection	One month	₱1,699.00
Electricity	One month	₱3,000.00
Food allowance	Project allowance	₱2,000.00
Operating and miscellaneous subtotal		**₱6,699.00**
TOTAL IMPLEMENTATION BUDGET	**One-month budget basis**	**₱55,199.00**

The laptop and phone amounts are retained from the proposed budget. The watch, internet and electricity amounts reflect the revised project estimates. The food allowance is treated as a single project expense.

Internet and electricity total ₱4,699.00 per month. For a longer implementation period, add ₱4,699.00 for each additional month. The one-month total does not include labor, repairs, store-publishing fees, or paid cloud/API usage beyond the budgeted no-cost allowances.

Zero-cost software entries describe the prototype budget, not unlimited service capacity or an open-source license for every product. Review provider quotas before expanding the system.

Technical references: Supabase pricing; Firebase pricing; Expo service costs; Expo Push Service cost; Gemini API pricing. Accessed 3 November 2026.

Software Specification

The system is to be implemented with the following software specifications:

Table 32
SOFTWARE SPECIFICATION AND HARDWARE SPECIFICATION

ElderCareAI: An AI-Driven IoT System for Elderly Monitoring and Caregiver Support SOFTWARE DEVELOPMENT REQUIREMENT SPECIFICATION	
Back-end:	
DBMS	PostgreSQL (Supabase)
Server-side Programming	TypeScript / Deno-compatible Supabase Edge Functions; PostgreSQL SQL functions
Cloud Service	Supabase Cloud (Auth, Realtime, Storage)
AI Service	Gemini API through server-side functions
Notification Service	Expo Push Service and Firebase Cloud Messaging v1
Front-end:	
Platform Technologies	Custom Android APK with native modules
Programming / Scripting Language	TypeScript / JavaScript
Software Framework	React Native, Expo
UI Environment	Android mobile phone
Administration Portal	HTML, CSS and JavaScript in a modern web browser
Watch and Health Integration:	
Watch Application	Kotlin, Android SDK and Wear OS
Live Sensor Integration	Samsung Health Sensor SDK; watch GPS for location
Health History Integration	Samsung Health Data SDK: sleep and recorded oxygen
Development Tools	Visual Studio Code, Android Studio, Android SDK, Gradle, JDK and Node.js

A custom APK supplies native health access and push support; Expo Go cannot load these modules. Configure Google sign-in, Samsung SDK access, backend secrets and Firebase/Expo credentials. Public Samsung SDK distribution has separate registration requirements. Retain compatible signing credentials for updates. Node.js is a development tool; Edge Functions use a Deno-compatible runtime.

Technical references: Expo custom native code; Supabase Edge Functions; Expo FCM v1 credentials. Accessed 3 November 2026.

Hardware Specification

The system is to be implemented with the following hardware specifications:

Component	Minimum Requirement	Recommended Requirement
Development CPU	Compatible 64-bit processor; virtualization support for emulator use	Recent Intel Core i5 / AMD Ryzen 5 class or better
Development Memory (RAM)	8 GB RAM for Android Studio without an emulator	16 GB for physical-device work; 32 GB for heavier emulator use
Development Storage	Space for Android Studio, SDKs, Gradle caches and project files	SSD with at least 32 GB free for tools, plus project and build space
Development Display	1280 × 800 pixels; USB or wireless debugging access	1920 × 1080 pixels or higher; stable internet and USB access
Development Operating System	64-bit Windows 10 supported by the tools	A currently supported 64-bit Windows release
Phone Operating System	Android 10 / API 29 or later	Android 13 or later, with current permission controls
Input Devices	Touchscreen controls on the Android phone and watch	Readable, accessible touch controls on both devices
Output Devices	Display, speaker and vibration	Display, sound and vibration with permitted Android push notifications
Wearable IoT Device	Samsung Galaxy Watch6 with the ElderCareAI Wear OS APK	Updated watch software and Health Sensor Service; enabled sensor and location permissions
Health and Location Sensing	Supported heart rate, SpO_2, wrist skin-temperature, motion and GPS readings	Valid sensor contact and GPS fix; sleep imported after recording and stress derived from HRV
Phone Health Integration	Samsung Health 6.30.2 or later and Galaxy Wearable on the paired phone	Patient's watch paired to the same phone; required sleep and oxygen read permissions granted
Bluetooth and Messaging	Watch–phone pairing; Google Play services on the phone for FCM	Stable pairing with notification registration and permission enabled
Cloud Server	Supabase Cloud with PostgreSQL	Supabase Authentication, Realtime, Storage and Edge Functions
Internet Connection	Internet access for uploads and cloud services	Stable Wi-Fi or mobile data for watch / phone synchronization and caregiver access
Power Supply	Rechargeable phone and smartwatch batteries with appropriate chargers	Routine charging; sufficient charge for active sensor and GPS use
Administrator Device	Laptop or desktop with a modern browser and internet connection	Secure, updated browser access to the administration portal

The development computer and runtime devices have different requirements, as identified in the component labels. Sensor updates and GPS depend on permissions, contact, signal and connectivity; active monitoring increases battery use. Other watch models require separate compatibility testing.

Technical references: Android Studio system requirements; Samsung Health Data SDK requirements. Accessed 3 November 2026.

Human Resource Specification

The research study, ElderCareAI, was designed to address the needs of the following users:

1. Caregiver

The caregiver manages the assigned older adult's profile, pairs the watch, reviews readings and location, maintains medications and appointments, responds to alerts, and consults Elle AI. The caregiver also enables notification permissions and checks whether displayed data is current.

2. System Administrator

The administrator uses the web portal to manage authorized user accounts and patient records, review stored monitoring data and audit activity, and generate system reports. Administrative privileges must be restricted to approved personnel.

3. Older Adult / Watch Wearer

The older adult wears the Galaxy Watch6, keeps it charged, and uses the watch's monitoring, spot oxygen, SOS and medication controls as needed. The caregiver can assist with setup and permissions. A separate caregiver-style phone account is not required simply to wear the paired watch.

Program Specification

The system is to be implemented with the following program specifications:

Functional Requirements

ElderCareAI is expected to perform the following functionalities:

- Authentication and profiles. Provide caregiver Google sign-in and separate authenticated administrator access. Maintain caregiver, patient and emergency-contact details with role-based access.
- Watch integration. Pair the Galaxy Watch6 with the correct patient, request the necessary permissions, display connection status and provide monitoring start/stop controls.
- Health monitoring. Display supported heart rate, oxygen, wrist skin temperature, steps, sleep and derived stress/HRV information with units and individual measurement times.
- Vital-event history. Save every accepted new vital event with its original time and source. Retain unacknowledged watch samples for retry and prevent duplicate retry records. Missing metrics remain blank rather than being fabricated.
- Location tracking. Display the patient's latest valid watch location and update the map when new positions arrive. Indicate location age or accuracy so an old fix is not presented as current movement.
- Medications and appointments. Create, update and display care schedules, instructions and reminders. Synchronize supported medication "Taken" actions between watch and app, with clear save and reminder-setup feedback.
- Threshold alerts. Evaluate valid uploaded readings against the configured app review rules. Save threshold warnings in Alerts and Notifications, apply duplicate/cooldown controls, and queue caregiver push delivery.
- SOS and notifications. Allow a watch SOS request to create a caregiver alert. Deliver SOS, vital warnings, medication reminders and appointment reminders through the configured notification paths. SOS is an assistance request, not an automatic emergency-services dispatch.
- Health history import. Import permitted sleep and recorded oxygen data from the selected patient watch through the phone's health history integration. Retry failed uploads without inventing or duplicating readings.
- Elle AI and analytics. Answer questions about saved patient history, summarize trends and provide clearly qualified statistical insights. Distinguish missing data from a confirmed negative result; avoid diagnosis or guaranteed health predictions.
- Reports. Provide patient vital-history and medication-schedule PDF exports, together with authorized administrative reporting.
- Administration and settings. Support account and record management, audit review, device status, notification preferences and sound settings. Expose clear permission, connectivity and synchronization errors.

Data Collection and Update Behavior

Data	Collection and Update Behavior
Heart rate	Updates from valid delivered watch sensor events while monitoring. New accepted readings are logged individually, including values above the warning threshold.
Blood oxygen (SpO_2)	Manual spot measurements and optional automatic spot checks while the watch app is in the foreground. Recorded overnight oxygen is imported when available; this is not a continuous background oxygen stream.
Wrist skin temperature	Uses valid supported skin-contact readings. This value is wrist skin temperature, not core body temperature.
HRV / stress level	Rolling RMSSD HRV is calculated from valid beat intervals over about 30–60 seconds. A personal reference supports an experimental 0–100 stress estimate; no score is invented while the baseline is insufficient.
Steps	Uses step-counter updates while monitoring. Time when monitoring is off cannot be assumed to have complete step coverage.
Sleep	Imports actual sleep sessions after they are available through the health-history SDK. It does not stream second by second or guarantee delivery immediately on waking.

Live updates are limited by sensor delivery, signal quality, connection state and upload success. Each vital retains its own measurement time. The stress estimate is experimental and not medical grade.

Technical references: Samsung sensor data and on-demand limitations. Accessed 3 November 2026.

Automatic Vital Review Rules

The following are the current application review rules. They are not individualized medical limits and should not be presented as a diagnosis or as proof that a patient is safe.

Metric	Warning Condition	Evaluation Basis
Heart rate	Below 50 BPM or above 110 BPM	A valid new recorded sample. Exactly 110 BPM does not meet the high-rate rule.
SpO_2	Below 95%	An accepted spot or imported oxygen reading; continuous coverage is not implied.
Sleep duration	Recorded session below 5 hours	Evaluated after the sleep record becomes available and is imported.
Wrist skin temperature	Change of at least 0.8°C AND at least 2 standard deviations	Requires at least five recent same-watch reference readings; uses up to seven readings within 24 hours.
HRV, stress and steps	No automatic threshold configured	Displayed and recorded where available; these values do not currently generate automatic threshold warnings.

A qualifying reading can create a warning after upload. Repeated notices for an active live rule are limited to one per hour; a return to range and a new crossing reset it. Imported records use deduplication. Push work is triggered after database commit, with scheduled recovery. Android and network conditions affect delivery.

Nonfunctional Requirements

ElderCareAI is designed to operate under the following parameters:

- Data quality. Validate values, timestamps and sensor quality. Mark missing or stale data clearly. Consumer wearable readings and derived scores must not be represented as clinical-grade measurements.
- Security and privacy. Use authenticated access, role checks, database row-level policies and encrypted network communication. Keep privileged backend, AI and push credentials out of client code.
- Reliability and integrity. Use event identifiers, acknowledgments and retries to reduce data loss and duplicate history. Preserve independent measurement times and source details.
- Responsiveness. Display new accepted data and queue threshold alerts promptly. Sensor intervals, continuous coverage and notification arrival times cannot be guaranteed.
- Background operation. Support Android push when the app is not open, using the correct APK, credentials, registration and permission. Force-stop, offline state, battery policies, muted channels and Do Not Disturb can affect delivery or sound.
- Usability and compatibility. Use readable labels, units, clear controls and understandable errors. Validate supported Android phones, Galaxy Watch6 and modern-browser administrator access.
- Maintainability and capacity. Keep mobile, watch and backend modules organized and testable. Monitor database growth, cloud quotas, failures and retry queues as the number of patients and recorded samples increases.

List of modules

The table below presents the proposed module assignments. Each proponent is responsible for implementing, testing and improving six modules of the updated ElderCareAI system.

An asterisk identifies use or participation, including wearing the watch. It does not grant permissions. Participation totals overlap when modules serve several user groups.

Table 31
LIST OF MODULES

Proponents	Modules	Elderly	Caregiver	Admin
	Authentication & User Management Modules			
Kristoffer James G. Layos	Landing page and role dashboards		*	*
	Login, authentication and role control		*	*
	Profiles and settings		*	*
	Caregiver account management			*
	Administrator access management			*
	Security and audit logs			*
No. of Points (1 point per module)		0	3	6
	Health Monitoring Modules			
John Wayne C. Planos	Watch pairing and sensor integration	*	*	
	Live vital signs and activity	*	*	
	Oxygen measurement and sleep history	*	*	
	HRV and experimental stress estimate	*	*	
	Medication and appointment care	*	*	
	Health history and PDF reports		*	*
No. of Points (1 point per module)		5	6	1
	Emergency & Monitoring Modules			
Melvence R. Bagonggong	SOS and caregiver response	*	*	
	Automatic vital-threshold alerts		*	*
	GPS tracking and map updates	*	*	
	Caregiver push and reminder notifications		*	*
	Elle AI and health analytics		*	
	System reports and device-status monitoring			*
No. of Points (1 point per module)		2	5	3

Total assigned modules: 18 (six per proponent). Participation totals: Elderly 7, Caregiver 14, Administrator 10.